राष्ट्रीय प्रौद्योगिकी संस्थान दिल्ली

NATIONAL INSTITUTE OF TECHNOLOGY DELHI

Department of Electrical Engineering

Project Synopsis

Student & Supervisor Details

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Course (Branch), Semester	**B. Tech - Electrical** **Semester- 4th**	**Project Code- EEPB 257**
Supervisor's Name	**Dr. Anmol Ratna Saxena**	

Project Details

Title of the Project	RFID enabled EV parking and charging system
Base Paper (Main Reference Paper)	"Smart Electric Vehicle Charging Station Management System using IoT and RFID Technology" – IEEE Research Papers related to Smart EV Charging and IoT Monitoring
Brief Introduction	The increasing use of Electric Vehicles (EVs) requires efficient charging infrastructure with proper parking and charging management. Existing charging stations often lack automation, real-time monitoring, and secure user authentication. This project proposes a Smart EV Charging Station Management System using ESP32, RFID, and IoT technology. RFID is used for user authentication and automatic slot allocation. IR sensors detect slot occupancy, while a servo motor controls gate operation. A relay module is used to simulate charging functionality. The system is integrated with Firebase Realtime Database for live cloud monitoring. Two web dashboards are developed: 1. Operator Dashboard – displays complete slot and charging information. 2. User Dashboard – displays assigned slot, charging status, battery percentage, and time remaining. The project demonstrates an IoT-based low-cost smart charging solution suitable for future smart city applications .
Motivation of Project	The increasing adoption of electric vehicles requires smart charging infrastructure capable of handling multiple users efficiently. Existing charging stations often lack

	automation, live monitoring, and proper slot allocation systems. Long waiting times and manual operations reduce user convenience. The motivation behind this project is: • To automate EV parking and charging operations. • To reduce human intervention in charging stations. • To provide real-time monitoring using IoT technology. • To improve user experience through live dashboards. • To develop a cost-effective smart charging prototype suitable for smart city applications. • To integrate RFID authentication for secure access control. This project aims to demonstrate how IoT and embedded systems can improve the efficiency and management of EV charging stations.
Related Work (Literature Review with reference)	Existing smart parking and EV charging systems use RFID, IoT, and cloud monitoring technologies for automation. Companies like Tesla and ChargePoint use real-time monitoring and automated charging management systems. ESP32-based systems are widely used because of their built-in WiFi capabilities. However, most existing systems provide either parking management or charging management separately. Integration of live slot allocation, charging monitoring, RFID authentication, and cloud dashboards in a single low-cost system remains limited.
Research Gap	From the literature survey, the following research gaps were identified: • Existing systems generally provide either smart parking or charging functionality separately. • Most systems do not include dedicated user and operator dashboards simultaneously. • Real-time slot allocation with live cloud updates is limited in low-cost prototypes. • Few systems integrate RFID authentication with charging management. • Many commercial systems are expensive and unsuitable for educational or prototype-level implementation. The proposed system addresses these limitations by developing a low-cost integrated smart EV charging station management solution using ESP32 and Firebase.
Objective	▪ To develop a smart EV charging and parking management system using ESP32 and IoT. ▪ To implement RFID-based user authentication and automatic slot allocation. ▪ To monitor slot occupancy and charging status in real time. ▪ To integrate Firebase cloud monitoring with user and operator dashboards. ▪ To provide a low-cost and automated EV charging solution
Tentative Work Plane/Methodology	The proposed system uses an ESP32 microcontroller integrated with RFID, IR sensors, servo motor, and relay module to automate EV charging station management. RFID cards are used for user authentication and automatic slot allocation. IR sensors continuously monitor slot occupancy and update the system in real time. When a vehicle occupies the assigned slot, charging simulation is activated

	through the relay module, while battery percentage, temperature, and charging time are monitored. The ESP32 sends live data to Firebase Realtime Database through WiFi connectivity. User and operator dashboards are developed using HTML, CSS, and JavaScript to display slot availability, charging status, and other live updates
References (In IEEE format)	[1] A. Kumar and S. Sharma, “IoT Based Smart EV Charging Station,” IEEE Conference, 2023. [2] M. Patel, “RFID Based Smart Parking and Charging System,” IEEE Access, 2022. [3] Espressif Systems, “ESP32 Technical Reference Manual,” 2024. [4] Firebase Documentation, Google, 2024. [5] NXP Semiconductors, “MFRC522 RFID Reader Datasheet,” 2023

Signature of student with date :

Signature of Supervisor with date :